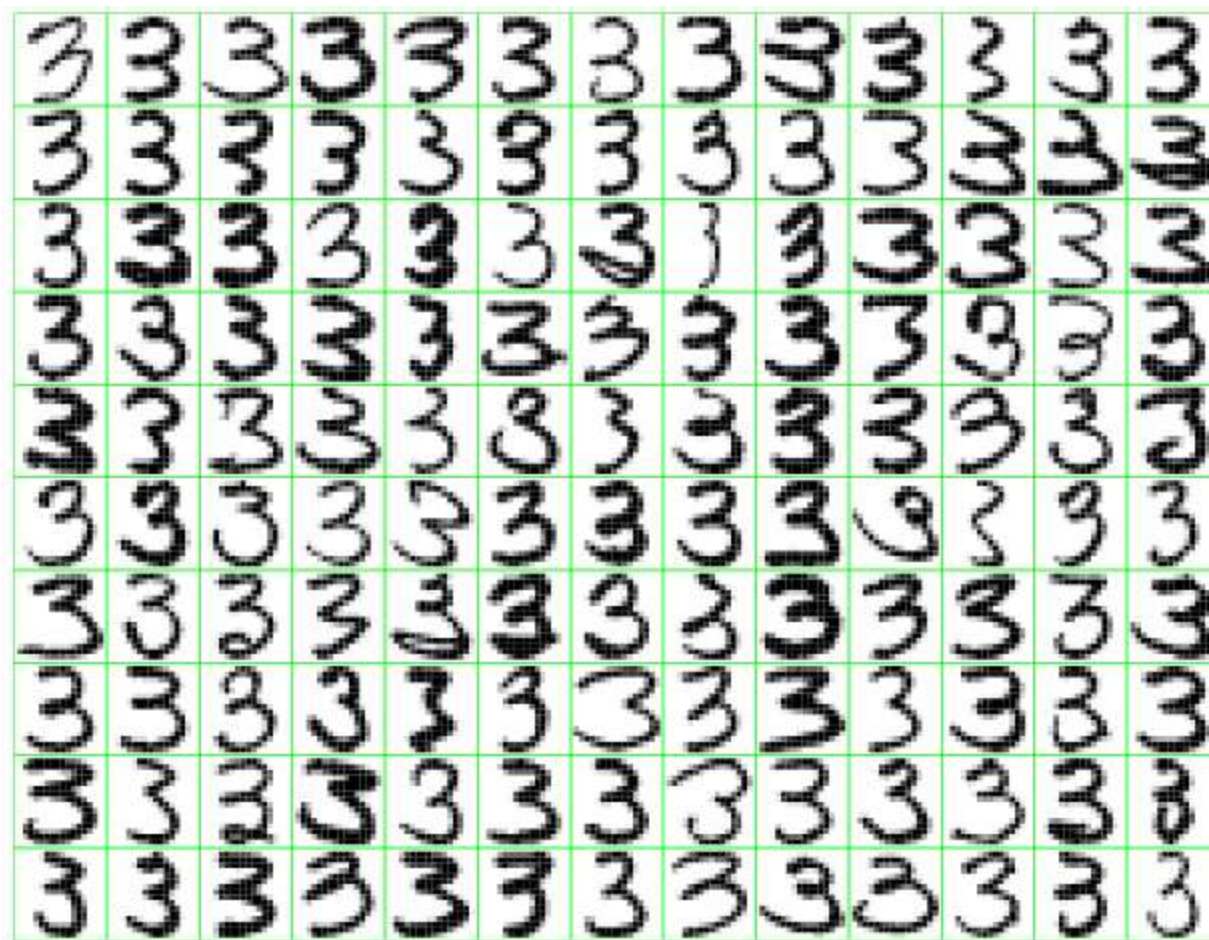


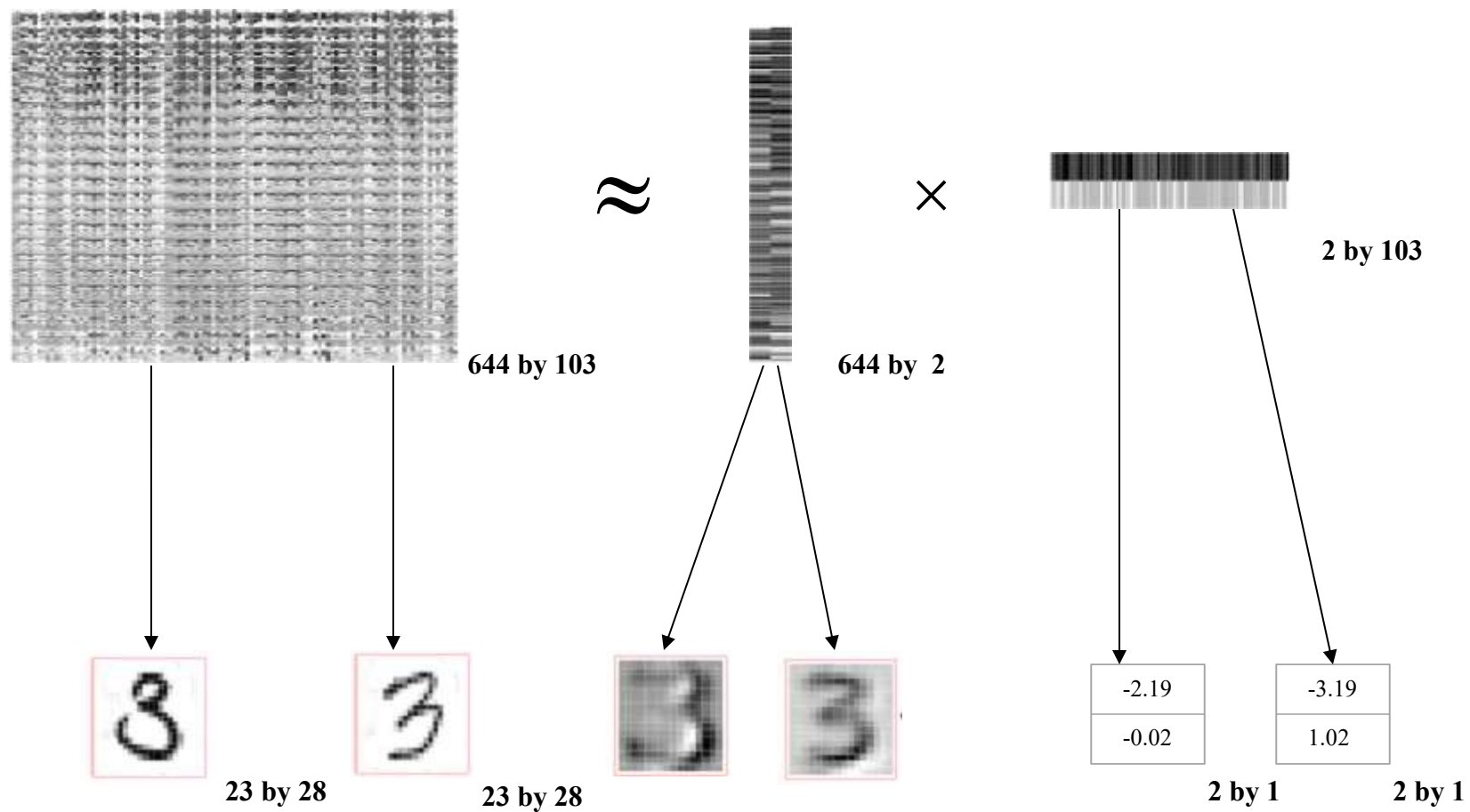
Lecture 3

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a method of dimensionality reduction/feature extraction that transforms the data from a d -dimensional space into a new coordinate system of dimension p , where $p \leq d$ (the worst case would be to have $p = d$).



A sample of 130 handwritten threes shows a variety of writing styles.



$$3 = \boxed{3} + 0.7 \cdot \boxed{3} + 0.2 \cdot \boxed{3}.$$

$$\begin{aligned}
 8 &= \boxed{3} + 0.7 \cdot \boxed{3} + 0.2 \cdot \boxed{3} . \\
 3 &= \boxed{3} + 0.3 \cdot \boxed{3} + 0.4 \cdot \boxed{3} .
 \end{aligned}$$

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 \end{aligned}$$

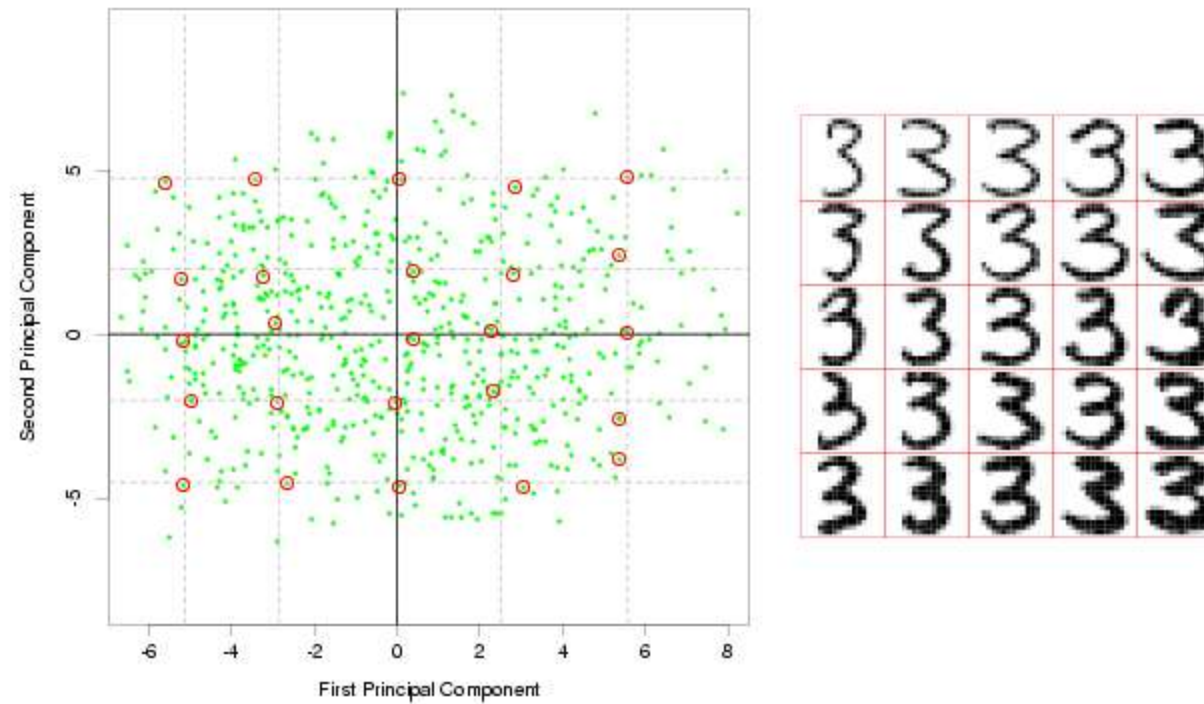
$$8 = \boxed{3} + 0.7 \cdot \boxed{3} + 0.2 \cdot \boxed{3}.$$

$$3 = \boxed{3} + 0.3 \cdot \boxed{3} + 0.4 \cdot \boxed{3}.$$

$$3 = \boxed{3} + 0.3 \cdot \boxed{3} + 0.7 \cdot \boxed{3}.$$

$$3 = \boxed{3} + 0.2 \cdot \boxed{3} + 0.4 \cdot \boxed{3}.$$

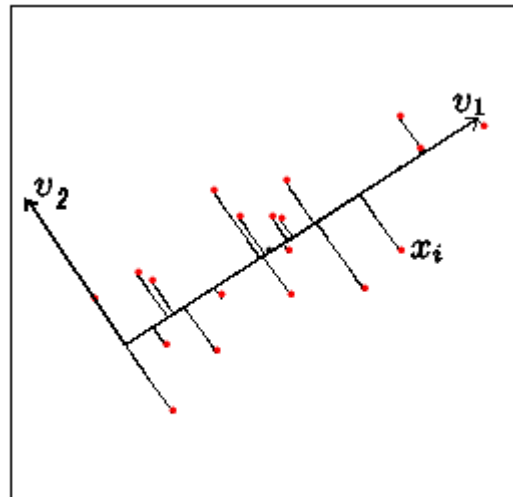
$$\begin{aligned} \hat{f}(\lambda) &= \bar{x} + \lambda_1 v_1 + \lambda_2 v_2 \\ &= \boxed{3} + \lambda_1 \cdot \boxed{3} + \lambda_2 \cdot \boxed{3}. \end{aligned}$$



Left plot: the first two principal components of the handwritten threes.

Right plot: the images corresponding to the red-circled points. These show the nature of the first two principal components.

Principal Component Analysis (PCA)



*The first and second principal component
of a set of data*

Principal Component Analysis (PCA)

- ▶ The goal is to preserve as much of the variance in the original data as possible in the new coordinate systems.
- ▶ Give data on d variables, the hope is that the data points will lie mainly in a linear subspace of dimension lower than d .
- ▶ In practice, the data will usually not lie precisely in some lower dimensional subspace.
- ▶ The new variables that form a new coordinate system are called **principal components** (PCs).

Principal Component Analysis (PCA)

- ▶ PCs are denoted by $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_d$.
- ▶ The principal components form a basis for the data.
- ▶ Since PCs are orthogonal linear transformations of the original variables there is at most d PCs.
- ▶ Normally, not all of the d PCs are used but rather a subset of p PCs, $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p$
- ▶ In order to approximate the space spanned by the original data points $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix}$ We can choose p based on what percentage of the variance of the original data we would like to maintain.

Principal Component Analysis (PCA)

The first PC, $\mathbf{u}_1$ is called **first principal component** and has the maximum variance, thus it accounts for the most significant variance in the data.

The second PC, $\mathbf{u}_2$ is called **second principal component** and has the second highest variance and so on until PC $\mathbf{u}_d$ which has the minimum variance.

Principal Component Analysis (PCA)

The most common definition of PCA, due to Hotelling is that, for a given set of data vectors $\mathbf{x}_i$, $i \in 1 \dots n$, the p principal axes are those orthonormal axes onto which the variance retained under projection is maximal.

Principal Component Analysis (PCA)

- ▶ In order to capture as much of the variability as possible, let us choose the first principal component, denoted by $\mathbf{u}_1$, to capture the maximum variance.
- ▶ Suppose that all centred observations are stacked into the columns of a $d \times n$ matrix X , where each column corresponds to a d -dimensional observation and there are n observations.
- ▶ The projection of n , d -dimensional observations on the first principal component $\mathbf{u}_1$ is $\mathbf{u}_1^T X$.

Principal Component Analysis (PCA)

We want projection on this first dimension to have maximum variance.

$$\text{var}(\mathbf{u}_1^T \mathbf{X}) = \mathbf{u}_1^T \mathbf{S} \mathbf{u}_1$$

where $\mathbf{S}$ is the $d \times d$ sample covariance matrix of $\mathbf{X}$.

- ▶ Clearly $\text{var}(\mathbf{u}_1^T X)$ can be made arbitrarily large by increasing the magnitude of $\mathbf{u}_1$.
- ▶ $\text{var}(\mathbf{u}_1^T X) = \mathbf{u}_1^T S \mathbf{u}_1$ where S is sample covariance matrix of sample data X .
- ▶ This means that the variance stated above has no upper limit and so we can not find the maximum.
- ▶ To solve this problem, we choose $\mathbf{u}_1$ to maximize $\mathbf{u}_1^T S \mathbf{u}_1$ while constraining $\mathbf{u}_1$ to have unit length.
- ▶ Therefore, we can rewrite the above optimization problem as:

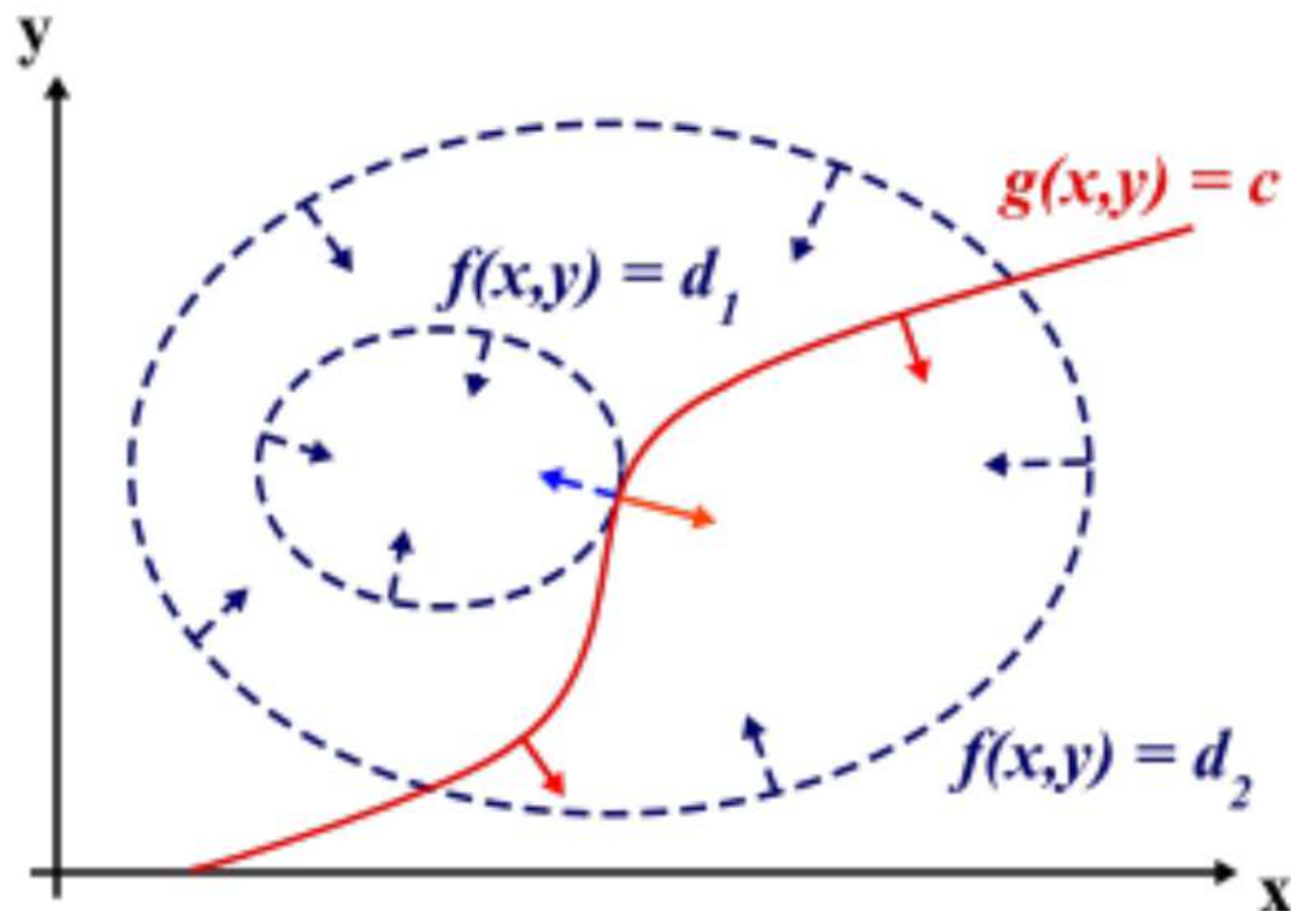
$$\begin{aligned} & \max \mathbf{u}_1^T S \mathbf{u}_1 \\ & \text{subject to } \mathbf{u}_1^T \mathbf{u}_1 = 1 \end{aligned}$$

To solve this optimization problem a Lagrange multiplier λ is introduced:

$$L(\mathbf{u}_1, \lambda) = \mathbf{u}_1^T S \mathbf{u}_1 - \lambda(\mathbf{u}_1^T \mathbf{u} - \mathbf{1})$$

Review of Lagrange Multiplier

Lagrange multipliers are used to find the maximum or minimum of a function $f(x, y)$ subject to constraint $g(x, y) = c$



"The red line shows the constraint $g(x,y) = c$. The blue lines are contours of $f(x,y)$. The point where the red line tangentially touches a blue contour is our solution." [Lagrange Multipliers, Wikipedia]

we define a new constant λ called a Lagrange Multiplier and we form the Lagrangian,

$$L(x, y, \lambda) = f(x, y) - \lambda g(x, y)$$

If $f(x^*, y^*)$ is the max of $f(x, y)$, there exists λ^* such that (x^*, y^*, λ^*) is a stationary point of L (partial derivatives are 0). In addition (x^*, y^*) is a point in which functions f and g touch but do not cross. At this point, the tangents of f and g are parallel or gradients of f and g are parallel, such that:

$$\nabla_{x,y} f = \lambda \nabla_{x,y} g$$

where, $\nabla_{x,y} f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) \leftarrow$ the gradient of f

$\nabla_{x,y} g = \left(\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y} \right) \leftarrow$ the gradient of g

Example :

Suppose we want to maximize the function $f(x, y) = x - y$ subject to the constraint $x^2 + y^2 = 1$.

We can apply the Lagrange multiplier method to find the maximum value for the function f ; the Lagrangian is:

$$L(x, y, \lambda) = x - y - \lambda(x^2 + y^2 - 1)$$

We want the partial derivatives equal to zero:

$$\frac{\partial L}{\partial x} = 1 + 2\lambda x = 0$$

$$\frac{\partial L}{\partial y} = -1 + 2\lambda y = 0$$

$$\frac{\partial L}{\partial \lambda} = x^2 + y^2 - 1$$

Solving the system we obtain two stationary points: $(\sqrt{2}/2, -\sqrt{2}/2)$ and $(-\sqrt{2}/2, \sqrt{2}/2)$. In order to understand which one is the maximum, we just need to substitute it in $f(x, y)$ and see which one as the biggest value. In this case the maximum is $(\sqrt{2}/2, -\sqrt{2}/2)$.

$$L(\mathbf{u}_1, \lambda) = \mathbf{u}_1^T S \mathbf{u}_1 - \lambda(\mathbf{u}_1^T \mathbf{u}_1 - 1) \quad (1)$$

Differentiating with respect to $\mathbf{u}_1$ gives d equations,

$$S \mathbf{u}_1 = \lambda \mathbf{u}_1$$

Premultiplying both sides by $\mathbf{u}_1^T$ we have:

$$\mathbf{u}_1^T S \mathbf{u}_1 = \lambda \mathbf{u}_1^T \mathbf{u}_1 = \lambda$$

$\mathbf{u}_1^T S \mathbf{u}_1$ is maximized if λ is the largest eigenvalue of S .

Clearly λ and $\mathbf{u}_1$ are an eigenvalue and an eigenvector of S .
Differentiating (??) with respect to the Lagrange multiplier λ gives us back the constraint:

$$\mathbf{u}_1^T \mathbf{u}_1 = 1$$

This shows that the first principal component is given by the eigenvector with the largest associated eigenvalue of the sample covariance matrix S .
A similar argument can show that the p dominant eigenvectors of covariance matrix S determine the first p principal components.

Note that the PCs decompose the total variance in the data in the following way :

$$\sum_{i=1}^d \text{Var}(\mathbf{u}_i^T X) = \mathbf{u}_i^T S \mathbf{u}_i = \lambda_i$$

$$= \sum_{i=1}^d (\lambda_i)$$

$$= \mathbf{Tr}(S)$$

$$= \text{Var}(X)$$

$Var(\mathbf{u}_i^T \mathbf{u}_i) = \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i = \lambda_i$ where λ_i is an eigenvalue of the sample covariance matrix $\mathbf{S}$ and $\mathbf{u}_i$ is its corresponding eigenvector.

$Var(\mathbf{u}_1^T \mathbf{X})$ is maximized if $\mathbf{u}_1$ is the eigenvector of $\mathbf{S}$ with the corresponding maximum eigenvalue.

Each successive PC can be generated in the above manner by taking the eigenvectors of $\mathbf{S}$ that correspond to the eigenvalues:

$$\lambda_1 \geq \dots \geq \lambda_d$$

such that

$$Var(\mathbf{u}_1^T \mathbf{X}) \geq \dots \geq Var(\mathbf{u}_d^T \mathbf{X})$$

Algorithm 1

Recover basis (PCs): Calculate $XX^T = \sum_{i=1}^n x_i x_i^T$ and let U = eigenvectors of XX^T corresponding to the top p eigenvalues.

Encode training data: $Y = U^T X$ where Y is a $p \times n$ matrix of encodings of the original data.

Reconstruct training data: $\hat{X} = UY = UU^T X$.

Encode test example: $y = U^T x$ where y is a p -dimensional encoding of x .

Reconstruct test example: $\hat{x} = Uy = UU^T x$.

Table: *Direct PCA Algorithm*

- ▶ A unique solution can be obtained by finding the singular value decomposition of X
- ▶ For each rank p , U consists of the first p columns of U .

$$X = U\Sigma V^T$$

- ▶ The columns of U in the SVD contain the eigenvectors of XX^T